

All-Depth Carrier Accessibility for Routed Composition

Factorization, Survivor Recursion, and Image–Kernel Obstructions

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Paper XX of the RIME program. This independently scoped theorem paper begins the post-protocol carrier/route line; it does not continue or reopen the SOF protocol sequence closed by Paper XV.

Abstract

Problem. Boolean support paths generally overapproximate nonzero routed operator compositions: consecutive supported blocks may lie on different invariant carriers, or a routed prefix may land in the kernel of the remaining suffix.

Approach. We consider a finite-dimensional vector space with a declared finite direct-sum carrier decomposition, a complete family of sector idempotents, and transport maps and sector projectors that preserve every carrier. Routed products are resolved carrierwise at every finite depth and at every internal cut.

Results. Every finite routed product factors over the declared carriers. Routed reachability therefore lies inside carrier-resolved support-path reachability; images and ranks decompose over common endpoint carriers; and disjoint endpoint carrier support forces exact vanishing at every finite depth. Vanishing at a cut is equivalent to carrierwise image–kernel containment, while a survivor recursion gives the complementary exact activity criterion. An exact four-dimensional Z_2 construction separates support from composition, and a complete one-carrier depth-two census records eight strict within-carrier obstructions among 24 supported candidates.

Boundary. Carrier factorization and image–kernel equivalence are standard linear algebra, and the all-length disjoint-carrier obstruction already appears in Paper III. Shared carrier support is necessary for possible activity but is not sufficient for a nonzero composition. The accompanying Rubik calculations are bounded numerical observations, not theorem premises, exact-zero certificates, or an all-depth classification.

Keywords: invariant direct sums; routed operator products; carrier support; image–kernel obstruction; Boolean reachability; sector projectors.

Notation Table

Symbol	Meaning
K, V	coefficient field and finite-dimensional vector space
B, V_b, Π_b	carrier labels, carrier summands, and carrier projections
I, Q_i	sector labels and complete sector idempotents
G, R_g	declared transport labels and carrier-preserving maps
$\text{Supp}_B(i)$	carrier support of sector i
$T_{i \leftarrow j}(\mathbf{g}; \mathbf{k})$	one labelled routed operator product
$\mathcal{C}_d(i, j)$	span of depth- d routed products with endpoints (j, i)
$\mathcal{P}_d^{(b)}, \mathcal{P}_d, \mathcal{R}_d$	carrier paths, aggregate support paths, and active routed pairs
A_r, B_r	suffix and prefix at an internal route cut
$W_r^{(b)}$	carrier-local propagated survivor subspace at step r
$D_{\text{act}}(i, j)$	minimum finite active depth, or ∞

1 Introduction

This paper develops a carrier-resolved formulation of finite-depth routed composition on a vector space with a declared direct-sum carrier decomposition, carrier-preserving transport maps, and carrier-preserving sector projectors.

The intended lift is

$$\text{two-step obstruction} + \text{carrier-resolved incidence} \implies \text{all-depth carrier accessibility}.$$

Here *accessibility* means exact activity of declared routed products, not Boolean graph reachability. It requires a carrier-local survival or image–kernel test after support admission.

The main theorem factors every finite routed product carrierwise. This separates support admission from exact activity: disjoint endpoint carrier support gives an all-depth zero obstruction, whereas carrier overlap leaves an exact image–kernel or survivor test inside each common carrier.

No Rubik-specific numerical fact or search convention enters the theorem. The contribution is the unified depth-indexed carrier formulation and its explicit separation of aggregate support paths, carrier-resolved paths, and routed operator products.

2 Related Work and Novelty Boundary

Paper III [3] already proves that a common block decomposition preserved by every transport and sector projector prevents any finite routed sequence from connecting sectors with disjoint endpoint block support (Theorem 7.2.2 and Corollary 4.4). It also gives the two-factor image–kernel criterion and a recursive longer-path test. Paper VII [2] develops the corresponding carrier-resolved zero-product locus, rank strata, and represented incidence profiles in its Section 6.

This paper does not reclaim those results. It places them in one depth-indexed routed-composition object and makes four additional pieces explicit:

1. the blockwise factorization of each complete routed product;
2. the containment from operator activity to carrier-resolved path support to aggregate support-path reachability;
3. the image–kernel criterion and survivor recursion at every internal cut, carrier by carrier; and
4. exact four-dimensional disjoint- and shared-carrier controls with a finite reproducibility package.

Thus Paper XX is a consolidation and extension of the Paper III–VII carrier/route line, not a replacement proof or a new claim to the underlying block-diagonal and zero-product facts.

Two ingredients are standard linear algebra: products of maps preserving a common finite direct-sum decomposition remain block diagonal, and $AB = 0$ if and only if $\text{im } B \subseteq \ker A$. This paper does not claim either ingredient as a new general theorem, nor does it claim that invariant carriers or sectorizations are canonical.

Standard treatments of invariant subspaces, direct-sum decompositions, and block matrices are given by Horn and Johnson [4]. The relation between matrix support patterns and directed graph structure belongs to combinatorial matrix theory [1]. Those backgrounds do not supply the depth-indexed typed route interface or the exact finite controls given here.

The contribution is the combined depth-indexed interface. Routed products retain one carrier through every factor; carrier-resolved path support remains distinct from operator activity

and aggregate graph support; the inherited endpoint obstruction becomes a direct factorization corollary; and shared-carrier activity is decided by an exact survivor recursion and cutwise image–kernel test. The exact finite controls exhibit both disjoint- carrier and strict within-carrier obstruction.

3 Carrier data

Let V be a finite-dimensional vector space over a field K . Fix a finite carrier index set B and a direct-sum decomposition

$$V = \bigoplus_{b \in B} V_b.$$

Write $\Pi_b : V \rightarrow V_b \hookrightarrow V$ for the associated projection. Thus

$$\Pi_b \Pi_c = \delta_{bc} \Pi_b, \quad \sum_{b \in B} \Pi_b = I_V.$$

Let G be a set of declared transport labels. For each $g \in G$, let $R_g \in \text{End}_K(V)$ satisfy

$$[R_g, \Pi_b] = 0 \quad \text{for every } b \in B.$$

Let I be a finite sector index set. A sectorization is a family of idempotents $Q_i \in \text{End}_K(V)$ satisfying

$$Q_i Q_j = 0 \quad (i \neq j), \quad \sum_{i \in I} Q_i = I_V,$$

and the carrier-preservation condition

$$[Q_i, \Pi_b] = 0 \quad \text{for every } i \in I, \quad b \in B.$$

Orthogonality or self-adjointness of the Q_i is not needed for the algebraic theorem. If the data are Hilbert-space projections, the same proof applies with orthogonal direct sums.

Define the carrier component of an operator by

$$R_g^{(b)} = \Pi_b R_g \Pi_b|_{V_b} \in \text{End}_K(V_b), \quad Q_i^{(b)} = \Pi_b Q_i \Pi_b|_{V_b} \in \text{End}_K(V_b).$$

The carrier support of sector i is

$$\text{Supp}_B(i) := \{b \in B : Q_i^{(b)} \neq 0\}.$$

This is an operator support label, not a statement that every vector in V_b belongs to sector i .

We identify

$$\bigoplus_{b \in B} \text{End}_K(V_b)$$

with the corresponding block-diagonal subalgebra of $\text{End}_K(V)$.

4 Routed products and depth spaces

For depth $d \geq 1$, choose transport labels

$$(g_1, \dots, g_d) \in G^d$$

and intermediate sectors $(k_1, \dots, k_{d-1}) \in I^{d-1}$. For endpoints $j, i \in I$, set

$$i_0 = j, \quad i_r = k_r \ (1 \leq r < d), \quad i_d = i,$$

and define the routed product

$$T_{i \leftarrow j}(\mathbf{g}; \mathbf{k}) = Q_{i_d} R_{g_d} Q_{i_{d-1}} R_{g_{d-1}} \cdots Q_{i_1} R_{g_1} Q_{i_0}.$$

For $d = 1$, the middle string is empty. For $d = 0$, define

$$T_{i \leftarrow j}(\emptyset) = Q_i Q_j.$$

The depth- d routed composition space is the finite-dimensional subspace

$$\mathcal{C}_d(i, j) = \text{span}_K \{T_{i \leftarrow j}(\mathbf{g}; \mathbf{k})\}.$$

The definition records actual operator products. A Boolean support path is only a candidate indexing a member of this family; it is not itself a nonzero composition witness.

Define the direct support relation by

$$j \longrightarrow i \iff Q_i R_g Q_j \neq 0 \text{ for some } g \in G.$$

Let $\mathcal{P}_d$ be the ordered endpoint relation defined by length- d paths in this support graph. Let $\mathcal{R}_d$ be the routed-composition relation

$$(j, i) \in \mathcal{R}_d \iff \mathcal{C}_d(i, j) \neq \{0\}.$$

Proposition 4.1 (Support Overapproximation).

For every $d \geq 1$,

$$\mathcal{R}_d \subseteq \mathcal{P}_d.$$

Proof. If one routed product is nonzero, none of its adjacent projected factors

$$Q_{k_r} R_{g_r} Q_{k_{r-1}}$$

can be zero. Each factor therefore supplies the corresponding edge of a length- d support path.

For each carrier b , define the carrier-restricted direct support relation by

$$j \longrightarrow_b i \iff Q_i^{(b)} R_g^{(b)} Q_j^{(b)} \neq 0 \text{ for some } g \in G,$$

and let $\mathcal{P}_d^{(b)}$ be its length- d path relation. The carrier factorization below gives the sharper three-level containment

$$\mathcal{R}_d \subseteq \bigcup_{b \in B} \mathcal{P}_d^{(b)} \subseteq \mathcal{P}_d.$$

The first inclusion is an all-depth operator statement; the second forgets the carrier label. A path in $\mathcal{P}_d$ can therefore be stitched from edges on different carriers and fail to represent one block-diagonal routed product.

The inclusion need not be equality. Theorem 5.1 below gives a structural sufficient condition for strictness: a support path whose endpoints have disjoint carrier support belongs to $\mathcal{P}_d \setminus \mathcal{R}_d$. $\square$

5 All-depth carrier factorization theorem

Theorem 5.1 (All-Depth Carrier Factorization).

Under the carrier data above, every routed product has the blockwise decomposition

$$T_{i \leftarrow j}(\mathbf{g}; \mathbf{k}) = \bigoplus_{b \in B} T_{i \leftarrow j}^{(b)}(\mathbf{g}; \mathbf{k}),$$

where

$$T_{i \leftarrow j}^{(b)}(\mathbf{g}; \mathbf{k}) = Q_i^{(b)} R_{g_d}^{(b)} Q_{k_{d-1}}^{(b)} \cdots Q_{k_1}^{(b)} R_{g_1}^{(b)} Q_j^{(b)} \in \text{End}_K(V_b).$$

Consequently,

$$\mathcal{C}_d(i, j) \subseteq \bigoplus_{b \in \text{Supp}_B(i) \cap \text{Supp}_B(j)} \text{End}_K(V_b).$$

In particular, if

$$\text{Supp}_B(i) \cap \text{Supp}_B(j) = \emptyset,$$

then

$$T_{i \leftarrow j}(\mathbf{g}; \mathbf{k}) = 0 \quad \text{for every finite depth } d \geq 0,$$

and hence $\mathcal{C}_d(i, j) = \{0\}$ for every d .

Proof. The commutation hypotheses imply that every transport and every sector projector is block diagonal for the direct sum $V = \bigoplus_b V_b$:

$$R_g = \bigoplus_{b \in B} R_g^{(b)}, \quad Q_\ell = \bigoplus_{b \in B} Q_\ell^{(b)}.$$

Products of block-diagonal operators are block diagonal, and multiplication of direct sums is componentwise. Substituting these decompositions into the definition of $T_{i \leftarrow j}$ gives

$$T_{i \leftarrow j} = \bigoplus_{b \in B} \left(Q_i^{(b)} R_{g_d}^{(b)} Q_{k_{d-1}}^{(b)} \cdots Q_{k_1}^{(b)} R_{g_1}^{(b)} Q_j^{(b)} \right),$$

which is the asserted factorization.

The same blockwise calculation proves the sharper inclusion above: if a routed product is nonzero, at least one carrier component is nonzero, and that component supplies a path in $\mathcal{P}_d^{(b)}$.

If $b \notin \text{Supp}_B(i)$, then $Q_i^{(b)} = 0$; if $b \notin \text{Supp}_B(j)$, then $Q_j^{(b)} = 0$. Therefore a component can be nonzero only when b lies in the intersection of the endpoint supports. This proves the containment and, when the intersection is empty, the all-depth zero conclusion. The depth-zero case is the same calculation with $T_{i \leftarrow j} = Q_i Q_j$.

Corollary 5.2 (Strict Support–Composition Separation).

If a length- d support path joins endpoints j, i with $\text{Supp}_B(i) \cap \text{Supp}_B(j) = \emptyset$, then

$$(j, i) \in \mathcal{P}_d \setminus \mathcal{R}_d.$$

□

Proof. The path gives membership in $\mathcal{P}_d$. Theorem 5.1 makes every routed product with those endpoints zero, so the pair is absent from $\mathcal{R}_d$. □

6 Rank and image consequences

The factorization gives more than a zero test.

Corollary 6.1 (Carrierwise Image Bound).

For every routed product,

$$\text{im } T_{i \leftarrow j} \subseteq \bigoplus_{b \in \text{Supp}_B(i) \cap \text{Supp}_B(j)} V_b.$$

Corollary 6.2 (Rank Additivity).

Over the declared field K ,

$$\text{rank } T_{i \leftarrow j} = \sum_{b \in B} \text{rank } T_{i \leftarrow j}^{(b)}.$$

Proof. The image of a block-diagonal map is the direct sum of the images of its carrier components. Dimensions of finite direct sums add, giving both claims.

For the depth spaces, define the carrier projection

$$\text{pr}_b : \mathcal{C}_d(i, j) \rightarrow \text{End}_K(V_b)$$

by taking the b -th block. Theorem 5.1 implies

$$\mathcal{C}_d(i, j) \subseteq \bigoplus_{b \in \text{Supp}_B(i) \cap \text{Supp}_B(j)} \text{pr}_b(\mathcal{C}_d(i, j)).$$

This is a carrierwise support envelope for routed composition. The block resolution of each operator is exact, but the displayed inclusion does not assert that every combination of carrierwise components is realized by one common route. □

6.1 Carrier profiles and minimum active depth

The **carrier profile** of sector i is the finite rank vector

$$\mathbf{p}_i = \left(\text{rank } Q_i^{(b)} \right)_{b \in B}.$$

Its nonzero coordinates recover $\text{Supp}_B(i)$. This vector distinguishes pure sectors, supported on one carrier, from hybrid sectors, supported on more than one carrier, while retaining the carrierwise sector dimensions discarded by a Boolean label.

Define the minimum active depth

$$D_{\text{act}}(i, j) = \inf\{d \geq 1 : \mathcal{C}_d(i, j) \neq \{0\}\},$$

with value ∞ when no finite active depth exists. This is the minimum **positive** active depth; the depth-zero identity-sector case is intentionally excluded.

Corollary 6.3 (Carrier and Graph Lower Gates).

If $D_{\text{act}}(i, j) < \infty$, then

$$\text{Supp}_B(i) \cap \text{Supp}_B(j) \neq \emptyset$$

and

$$\text{dist}_{\mathcal{G}}(j, i) \leq D_{\text{act}}(i, j).$$

Proof. The first condition is the contrapositive of Theorem 5.1. The second follows from Proposition 4.1 at the first active depth.

Neither gate is sufficient. Shared carrier support can still fail the image–kernel criterion, and a shortest graph path can be operator-inactive. A finite census through depth N therefore records only $D_{\text{act}}(i, j) \leq N$ or “not observed through N ”; it cannot replace ∞ by a finite-search null value. $\square$

7 Arbitrary-depth image–kernel obstruction

Every internal cut of a route supplies an exact obstruction criterion. For a depth- d route and $1 \leq r < d$, write

$$B_r = Q_{i_r} R_{g_r} Q_{i_{r-1}} \cdots Q_{i_1} R_{g_1} Q_{i_0}$$

for the routed prefix, and write

$$A_r = Q_{i_d} R_{g_d} Q_{i_{d-1}} \cdots R_{g_{r+1}} Q_{i_r}$$

for the routed suffix. Then the full routed product is $A_r B_r$. Under the carrier hypotheses, both factors preserve every V_b ; define their carrier blocks by

$$A_r^{(b)} := A_r|_{V_b} \in \text{End}_K(V_b), \quad B_r^{(b)} := B_r|_{V_b} \in \text{End}_K(V_b).$$

Theorem 7.1 (Arbitrary-Depth Image–Kernel Criterion).

At every internal cut r ,

$$T_{i \leftarrow j}(\mathbf{g}; \mathbf{k}) = 0 \iff \text{im } B_r \subseteq \ker A_r.$$

Under the carrier hypotheses of Theorem 5.1, this is equivalent to

$$\text{im } B_r^{(b)} \subseteq \ker A_r^{(b)} \quad \text{for every } b \in B.$$

Proof. For arbitrary linear maps A_r and B_r , the product $A_r B_r$ is zero exactly when A_r annihilates every vector in the image of B_r . This proves the first equivalence. Theorem 5.1 makes both prefix and suffix block diagonal, so $A_r B_r = \bigoplus_b A_r^{(b)} B_r^{(b)}$. A direct-sum operator is zero exactly when every component is zero; applying the first equivalence on each V_b gives the carrierwise statement. $\square$

7.1 Carrierwise survivor recursion and promotion

The cut criterion separates the structural obstruction from the within-carrier promotion problem. For every carrier b , define

$$W_0^{(b)} = \text{im } Q_j^{(b)}, \quad W_r^{(b)} = Q_{i_r}^{(b)} R_{g_r}^{(b)} W_{r-1}^{(b)} \quad (1 \leq r \leq d).$$

Theorem 7.2 (Carrierwise Survivor Recursion and Promotion).

For every declared route,

$$\text{im } T_{i \leftarrow j}(\mathbf{g}; \mathbf{k}) = \bigoplus_{b \in B} W_d^{(b)}.$$

Consequently,

$$T_{i \leftarrow j}(\mathbf{g}; \mathbf{k}) \neq 0 \iff W_d^{(b)} \neq \{0\} \text{ for at least one } b \in B.$$

Proof. The restriction of the routed product to V_b is the carrier-local product from Theorem 5.1. Its image is obtained by starting with $\text{im } Q_j^{(b)}$ and applying the carrier-local factors in route order, which is precisely the recursion defining $W_d^{(b)}$. The full product is the direct sum of these restrictions, so its image is the direct sum of their images. A finite direct sum is nonzero exactly when at least one summand is nonzero.

At each step this gives the exact local survival gate

$$W_r^{(b)} \neq \{0\} \iff W_{r-1}^{(b)} \not\subseteq \ker(Q_{i_r}^{(b)} R_{g_r}^{(b)}).$$

Equivalently, fix one route, cut, and carrier b . Set

$$A_b = A_r^{(b)}, \quad B_b = B_r^{(b)}.$$

Then

$$T_{i \leftarrow j}^{(b)} = A_b B_b = 0 \iff \text{im } B_b \subseteq \ker A_b.$$

Thus the all-depth theory has the precise boundary:

$\text{Supp}_B(i) \cap \text{Supp}_B(j) = \emptyset \implies T_{i \leftarrow j} = 0$ at every finite depth,
 $\text{Supp}_B(i) \cap \text{Supp}_B(j) \neq \emptyset \implies$ an exact survivor/image–kernel test is still required.

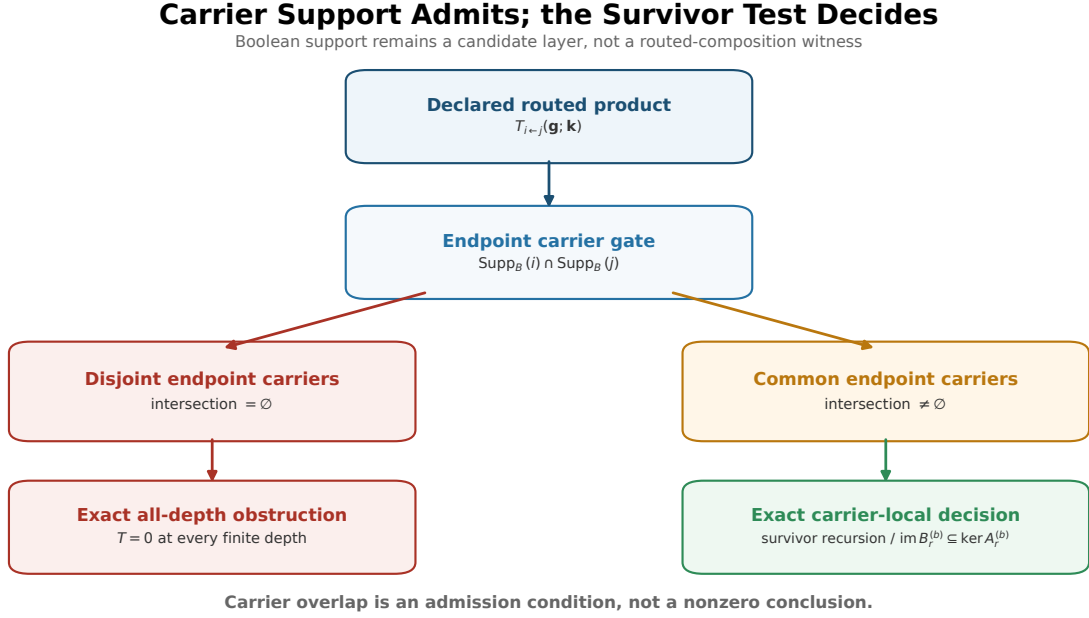


Figure 1. Endpoint carrier support supplies an exact all-depth obstruction only in the disjoint case. Carrier overlap admits a route to the carrier-local survivor or image–kernel test; it does not establish a nonzero composition.

An intermediate sector can carry edges incident to different carriers. That incidence does not mix carriers, because every inserted operator is block diagonal. A graph route may therefore exist while every carrier component of the routed product vanishes.

The shared-carrier promotion condition is strict rather than heuristic: the full route is active exactly when at least one carrier violates the image–kernel containment. In particular, it is sufficient that, for one carrier b , $B_b \neq 0$ and A_b is injective on $\text{im } B_b$. It is also sufficient that $\text{im } B_b = \text{im } Q_{k_r}^{(b)}$ and $A_b \neq 0$. $\square$

8 Sharpness and examples

At the level of endpoint carrier support alone, disjointness gives the strongest unconditional vanishing conclusion available here. If the endpoints share a carrier b , the theorem only says that the product may live on V_b ; it does not force it to vanish. For example, with one carrier $B = \{b\}$ and $Q_i = Q_j = R_g = I$, every depth product is I .

Conversely, shared carrier support is not sufficient for nonzero composition. Take $V_b = K^2$, let Q_j project onto the first coordinate and let the leftmost factor in a route project onto the second coordinate. The endpoint supports both equal $\{b\}$, but the routed product is zero by image–kernel containment.

The theorem is therefore sharp at the level claimed: endpoint carrier disjointness is sufficient for all-depth vanishing, while carrier overlap is only a necessary gate for possible activity.

Proposition 8.1 (Exact Four-Dimensional Z2 Witness).

There is a four-dimensional representation of Z_2 and a three-sector decomposition for which $\mathcal{R}_2 \subsetneq \mathcal{P}_2$, while the missing endpoint pair is inactive at every finite depth.

8.1 Construction and proof

Let $V = A \oplus B$ with ordered basis (a_0, a_1, b_0, b_1) and let the nonidentity element $s \in Z_2$ act by

$$R_s(a_0) = a_1, \quad R_s(a_1) = a_0, \quad R_s(b_0) = b_1, \quad R_s(b_1) = b_0.$$

Let Q_0, Q_1, Q_2 be the coordinate projectors onto the sectors

$$\text{im } Q_0 = \text{span}\{a_0\}, \quad \text{im } Q_1 = \text{span}\{a_1, b_0\}, \quad \text{im } Q_2 = \text{span}\{b_1\}.$$

Then

$$Q_1 R_s Q_0 \neq 0, \quad Q_2 R_s Q_1 \neq 0,$$

so the support graph contains $0 \rightarrow 1 \rightarrow 2$. However,

$$Q_2 R_s Q_1 R_s Q_0 = 0.$$

Indeed, the first step lands in $Ka_1 \subset A$, and every later factor preserves $A \oplus B$, while Q_2 is supported only on B . More strongly, Theorem 5.1 gives $\mathcal{C}_d(2, 0) = \{0\}$ for every d . This is an exact finite witness of strict support–composition inclusion; no numerical tolerance is involved in the statement or proof.

8.2 Exact shared-carrier obstruction census

A second four-dimensional control uses one carrier $V_b = K^4$, three coordinate sectors of dimensions $(1, 2, 1)$, and two permutation transports

$$g = (0 \ 1), \quad h = (2 \ 3).$$

Every nonzero sector has the same carrier support $\{b\}$. Nevertheless,

$$Q_1 R_g Q_0 \neq 0, \quad Q_2 R_h Q_1 \neq 0, \quad Q_2 R_h Q_1 R_g Q_0 = 0.$$

The incoming image is Ke_1 , whereas the outgoing factor is supported on the e_2 direction inside the same intermediate sector. This is strict within-carrier image–kernel containment, not a disjoint-endpoint obstruction.

The exact census enumerates all $3^3 \cdot 2^2 = 108$ depth-two labelled routes in this model. It finds 24 routes with two nonzero adjacent factors, of which 16 have nonzero products and eight are strict shared-carrier obstructions. Every matrix entry and rank is evaluated over integers; no tolerance is used. This finite result is a **Computational Certificate** for the declared model, not a frequency or genericity theorem.

9 Finite Evidence Scope and Cross-Paper Boundary

A finite census is interpretable only after separately declaring:

1. the carrier decomposition and exact carrier-support rule;
2. the finite generator and sector sets;
3. the routed depth range and route enumeration policy;

4. the within-carrier product or image–kernel test;
5. the coefficient domain, numerical tolerance, and exactness status.

The theorem does not validate a numerical sector registration. It only says what follows once the carrier-preservation hypotheses and the carrier labels are established.

Relation to Paper XXI. Paper XXI studies a specific marked projective finite-field route family by pole/preimage arithmetic. The two papers belong to the same carrier/route research line, but no theorem dependency is claimed at present. In particular, Paper XXI is not a corollary of the carrier factorization proved here: it does not currently register a decomposition $V = \bigoplus_b V_b$ and verify that its labelled generators and marked-sector projectors preserve every declared carrier. No cross-paper implication follows without those data and explicit verification of the carrier hypotheses.

10 Claim Status and Boundary

The accompanying evidence consists of an exact shared-carrier census, finite control censuses, and a Rubik depth-two image–kernel audit. The shared-carrier census uses integer permutation and projector matrices. Statements obtained from `complex128` norms remain bounded numerical observations, including the observation that the audited Rubik products lie below the declared tolerance. They do not establish exact projected zero or an all-depth Rubik classification. The exact Z_2 all-depth conclusion follows from the displayed construction and Theorem 5.1, not from its floating-point replay.

No claim is made that a carrier decomposition is canonical, that Boolean graph support implies an active routed composition, that a bounded null establishes all-depth vanishing, or that shared carrier support alone guarantees a nonzero product.

Statement	Status	Provenance or role
Carrierwise factorization of every finite routed product	Theorem	explicit all-depth routed formulation of the common-block argument
All-depth zero for disjoint endpoint carrier support	Theorem	antecedent in Paper III; recovered here as a factorization corollary
Routed activity lies inside carrier-path and support-path reachability	Proposition	depth-indexed three-level containment
Disjoint-carrier graph path gives strict inclusion	Corollary	consequence of factorization plus a declared support path
Cutwise carrier image–kernel criterion	Theorem	arbitrary-cut formulation of the standard zero-product test
Carrierwise survivor recursion and exact promotion	Theorem	all-depth shared-carrier activity criterion
Exact four-dimensional Z_2 separation witness	Proposition	new exact finite witness in this paper
Exact one-carrier obstruction census	Computational Certificate	8 strict obstructions among 24 supported candidates
Carrierwise image bound	Corollary	direct factorization consequence
Rank additivity across carriers	Corollary	standard direct-sum consequence
Carrier profile and minimum active depth	Definition	paper-local typed summary
Shared carriers imply nonzero routed composition	False in general	excluded converse
Graph path implies nonzero routed composition	False in general	excluded promotion
Rubik 228-dimensional census	Computational Observation	bounded evidence; not theorem proof
Higher-depth Rubik accessibility classification	Research Program	open application problem

11 Conclusion

A declared carrier decomposition resolves each routed sector composition into within-carrier products. In the Paper III–VII line this recovers the exact all-depth zero obstruction for disjoint endpoint carriers, while the present depth-indexed formulation also exposes carrier-resolved path containment, survivor recursion, and the image–kernel test at every cut. Shared carrier support is not a conclusion but an admission gate: exact activity occurs only when at least one carrier-local propagated image survives. Boolean support paths are therefore candidate routes rather than composition witnesses. Exact applications must establish carrier preservation and zero tests in their declared coefficient domain; bounded numerical applications must retain their tolerance and evidence status.

Appendix A Computational Artifacts

The default directory is `experiments/paper20/`. The A7 receipt records exact artifact paths and digests; its exact receipt and closure digests and validation boundary are indexed in `README.md`. The listed artifacts are available in the [RIME repository](#). They support the finite evidence surfaces and bind the local release closure; they do not replace the manuscript proofs.

Artifact	Role	Short path
A1	exact shared-carrier producer	<code>within_carrier_census.py</code>
A2	exact shared-carrier result	<code>results/within_carrier_obstruction_v1.json</code>
A3	exact shared-carrier validator	<code>validate_within_carrier_census.py</code>
A4	bounded carrier census producer and validator	<code>census.py</code> , <code>validate_results.py</code>
A5	Rubik depth-two image–kernel bundle	<code>image_kernel_census.py</code> , <code>results/image_kernel/</code>
A6	Rubik image–kernel validator	<code>validate_image_kernel.py</code>
A7	release manifest and local closure receipt	<code>release-manifest.json</code> , <code>results/carrier_accessibility_v1.release-receipt.json</code>

The A7 receipt performs **local closure verification**, not independent validation. Its ordered artifact closure excludes the receipt itself. The manifest is an upstream closure declaration and therefore does not contain the receipt digest; the exact receipt SHA-256 is recorded only in the downstream README index. Thus no artifact in the receipt closure depends on the receipt as a premise.

References

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